6 Startup

6.1 Setting the nominal current

Determine the nominal current of the load prior to startup. Set this for the relevant channel.

For the 1-channel and 4-channel circuit breakers, the nominal current is set (programmed) via the LED button; for the 2-channel circuit breakers, it is set via the rotary switch.

Note:
- All channels are switched off by default.
- The channel can be switched on and off via the channel LED button.
- The previous channel states are restored when the device is switched on.
- The channels in the system start cascaded with a standard delay of 50 ms.

6.1.1 Programming (1-channel and 4-channel circuit breakers)

It is only possible to program adjustable circuit breakers (designation in the name, e.g., 1-10A).

- Start programming mode by pressing the LED button (>2 seconds). The LED shows the nominal current set via a flashing yellow rhythm (e.g., 4 flashes for 4 A).
- Set the required nominal current by repeatedly pressing the LED button. Example: Press 3 times for 3 A, this will then be shown by 3 flashes.
- Press the channel LED button for >2 seconds to save the new current value.

After 60 seconds without activity, programming mode automatically switches off.

Note: Initial programming

After the channel has been switched on, the channel may switch off and the LED may flash red.
- Check the currents that have been set.

Visual signaling

Table 6-1 1-channel and 4-channel visual signaling

Signaling (LED)		Description
Off	LED off	Channel switched off
Green	On	Channel switched on
Yellow	On	Channel switched on, channel load >80% of the nominal current set. → Check the configuration.
	Flashing	Channel is in programming mode
	Flashes twice	Check the installation, no communication with power module. → Check the lateral contact surfaces and metals for damage.
Red	On	Channel switched off, overload or short-circuit tripping, 5-second cool-down phase. → It is not possible to switch the channel on while the LED is illuminated.
	Flashing	Channel switched off, overload or short-circuit tripping. → Switch the channel back on by pressing the corresponding LED button.
Red/yellow	Flashing	Channel is in overload mode and will be switched off in approx. 5 seconds. → Check the configuration.

6.1.2 Programming (2-channel circuit breakers)

- Switch the channel on by pressing the corresponding LED button.
- Set the necessary nominal current via the rotary switch. The channel LED starts to flash green.
- Press the channel LED button for >2 seconds to save the new current value.

Note: “RC” switch position

The “RC” position of the rotary switch stands for “Remote Control”. If you use a power module with a bus interface, such as IO-Link or Profinet, you activate the option of controlling the nominal current of the channel via the power module.

Note: Initial programming

After switching on the channel, the channel may switch off due to an overload or short circuit and the LED may flash red.

– Set the nominal current via the rotary switch while it is switched off. The LED now flashes red/green.
– Press the channel LED button for >2 seconds to save the new current value.

Nominal current assistant

- Program the channel to the highest value (e.g., 10 A).
- Start up the system so that the operating current flows.
- Turn the rotary switch down step-by-step so that it approaches the operating current that is currently flowing. This will enable you to find the appropriate setting for the channel. The channel LED flashes green in the process. If the channel LED changes and starts to flash yellow/green, the selected setting is too low for the system current that is currently flowing.
- Turn the rotary switch back up one position.
- Press the channel LED button for >2 seconds to save the new current value.

Visual signaling

Table 6-2 2-channel visual signaling

Signaling (LED)		Description
Off	LED off	Channel switched off
Green	On	Channel switched on
	Flashing	Channel switched on, programming mode is active, and the nominal current setting differs from the saved value. → To save the set nominal current, press the channel LED button for 2 seconds or reset the rotary switch to the previous value. The previous value has been set again once the LED stops flashing.
Yellow	On	Channel switched on, channel load >80% of the nominal current set. → Check the configuration.
	Flashes twice	Check the installation, no communication with power module. → Check the lateral contact surfaces and metals for damage.
Red	On	Channel switched off, overload or short-circuit tripping, 5-second cool-down phase. → It is not possible to switch the channel on while the LED is illuminated.
	Flashing	Channel switched off, overload or short-circuit tripping. → Switch the channel back on by pressing the corresponding LED button.
Red/green	Flashing	Channel switched off, programming mode is active, and the nominal current was adjusted after a shutdown caused by a fault in the rotary switch. → Save the newly set nominal current by pressing the channel LED button for 2 seconds or reset the rotary switch to the previous value. The previous value has been set again once the LED flashes only red.
Yellow/red	Flashing	Channel is in overload mode and will be switched off in approx. 5 seconds. → Check the configuration.

6.1.3 Locking the nominal current setting

Lock the nominal current setting using the power LED button (PWR) on the power module.

Setting the programming lock

- Press the power LED button for 3 seconds.

The programming lock is set for all the circuit breaker modules connected. The nominal current can no longer be modified. The power LED flashes yellow three times.

Resetting the programming lock

- Press the power LED button for 3 seconds while the programming lock is active.

The programming lock will be deactivated for all the circuit breaker modules connected. The nominal current can be modified again. The power LED flashes green three times.

6.2 Connecting the loads

Switching off the channel

Before connecting the load, make sure that the relevant channel on which the load is to be operated is switched off (LED is off).

Connecting the load

Connect the load via the load output of the relevant circuit breaker module as described in Section 5, “Connecting or removing cables”.

Switching on the channel

Then start the channel up again by pressing the channel LED (green LED).

6.3 Diagnostic and status indicators

6.3.1 Indicators on the power module

The power modules have status LEDs to indicate the current operating state.

Table 6-3 Visual signaling of the EtherNet/IP™ power module (PM EIP)

LED Designation	Color	Meaning	Status	Description
PWR	Green/yellow/red	Voltage indication	Green on	Operating voltage present
			Red on	Operating voltage outside the nominal voltage range
			Off	Operating voltage not present
NET	Green/red	EtherNet/IP™ connection	Flashing red/green	Self-test following power supply switch-on
			Green on	CIP/EIP connection active
			Red on	Duplicate IP address detected
			Flashing green	IP address available, no CIP/EIP connection
			Off	No voltage is applied to the device or no IP address
MOD	Green/red	EtherNet/IP™ connection	Flashing red/green	Self-test following power supply switch-on
			Green on	Device OK
			Red on	EIP: “Major unrecoverable fault”
			Flashing green	No device configuration
			Flashing red	EIP: “Major recoverable fault”
			Off	No voltage is applied to the device
RDY	Green	Ready	Green on	Device is ready for operation
			Off	Device is not ready for operation

Table 6-4 Visual signaling of the power module with status-reset (PM S-R)

LED Designation	Color	Meaning	State	Description
PWR	Green/yel-low/red	Voltage indicator	Green on	Operating voltage present
			Red on	Operating voltage outside the nominal voltage range
			Off	Operating voltage not present
System LED	Green	Indication of sys-tem state	Green on	System state OK
	Yellow		Yellow on	At least one channel has reached 80% of the nominal current
	Red		Red on	At least one channel has tripped due to an error

Table 6-5 Signal connections of the CAPAROC PM S-R

Identification	Designation	Description
S	Status	0 V = at least one channel has tripped due to an error 12 V ... 24 V = system without errors
RST	Reset	A falling edge (change from 12 V ... 24 V to 0 V) resets all the channels tripped
80%	80%	0 V = system without errors 12 V ... 24 V = at least one channel has more than 80% of the nominal current set

6.3.2 Indicators on the protective device

The protective device has one status LED per channel to indicate the current operating state of the channel.

For the signaling of the operating state and the relevant status, please refer to Section “Programming (1-channel and 4-channel circuit breakers)” on page 39 and Section “Programming (2-channel circuit breakers)” on page 40.

7 Process, parameter, and diagnostic data

7.1 Structure of the output assembly

The channel states of each circuit breaker module can be controlled via a byte in the output assembly.

Table 7-1 Output assembly

Byte offset	Name	EIP data type	Data size	Default	Read/write
0	Error reset	Byte	1	0	Write
1	Set channel status of module 1	Byte	1	0	Write
2	Set channel status of module 2	Byte	1	0	Write
3	Set channel status of module 3	Byte	1	0	Write
4	Set channel status of module 4	Byte	1	0	Write
5	Set channel status of module 5	Byte	1	0	Write
6	Set channel status of module 6	Byte	1	0	Write
7	Set channel status of module 7	Byte	1	0	Write
8	Set channel status of module 8	Byte	1	0	Write
9	Set channel status of module 9	Byte	1	0	Write
10	Set channel status of module 10	Byte	1	0	Write
11	Set channel status of module 11	Byte	1	0	Write
12	Set channel status of module 12	Byte	1	0	Write
13	Set channel status of module 13	Byte	1	0	Write
14	Set channel status of module 14	Byte	1	0	Write
15	Set channel status of module 15	Byte	1	0	Write
16	Set channel status of module 16	Byte	1	0	Write
17	Reserved	Byte	1	0	Write
18	Reserved	Byte	1	0	Write
19	Reserved	Byte	1	0	Write

The individual data types listed in this table are explained in the following subsections.

7.1.1 Reset byte

Using the global reset, the command to reset all triggered channels can be sent via the output assembly. Once the bit is set, it is ensured that the command is sent once.

For another reset command, first set the bit to “0” and then to “1” again.

Table 7-2 Reset byte

Error reset byte	Bit	Function	Description
Byte 0	0	Error reset	0 = No change
			1 = Switch triggered channels back on
	1 ... 7	Reserved	

7.1.2 Channel states per module

The channel states of each circuit breaker module can be controlled via a byte in the output assembly.

The structure of the control byte for each module is as follows:

Table 7-3 Channel states per module

Set byte of channel status of module X	Bit	Function	Value range
Byte 0	0	Channel 1, status	0 = Off 1 = On
	1	Channel 2, status	0 = Off 1 = On
	2	Channel 3, status	0 = Off 1 = On
	3	Channel 4, status	0 = Off 1 = On
	4 ... 6	Reserved	
	7	Release bit	0 = Data is not applied 1 = Data is applied

7.2 Structure of the input assembly

The channel states of each circuit breaker module can be read out via a byte in the input assembly.

The structure of the input assembly is as follows:

Table 7-4 Structure of the input assembly

Byte offset	Description	EIP data type	Data size	Default	Read/write
0	Global status	USINT	1	0	Read
1	Global module counter	USINT	1	0	Read
2 ... 3	Global total current	INT	2	0	Read
4 ... 5	Global input voltage	INT	2	0	Read
6	Module 1, channel 1, status byte	USINT	1	0	Read
7	Module 1, channel 1, nominal current	USINT	1	0	Read
8	Module 1, channel 1, currently flowing current	USINT	1	0	Read
9	Module 1, channel 2, status byte	USINT	1	0	Read
10	Module 1, channel 2, nominal current	USINT	1	0	Read
11	Module 1, channel 2, currently flowing current	USINT	1	0	Read
12	Module 1, channel 3, status byte	USINT	1	0	Read
13	Module 1, channel 3, nominal current	USINT	1	0	Read
14	Module 1, channel 3, currently flowing current	USINT	1	0	Read
15	Module 1, channel 4, status byte	USINT	1	0	Read
16	Module 1, channel 4, nominal current	USINT	1	0	Read
17	Module 1, channel 4, currently flowing current	USINT	1	0	Read
18	Module 2, channel 1, status byte	USINT	1	0	Read
19	Module 2, channel 1, nominal current	USINT	1	0	Read
20	Module 2, channel 1, currently flowing current	USINT	1	0	Read
21	Module 2, channel 2, status byte	USINT	1	0	Read
22	Module 2, channel 2, nominal current	USINT	1	0	Read
23	Module 2, channel 2, currently flowing current	USINT	1	0	Read
24	Module 2, channel 3, status byte	USINT	1	0	Read
25	Module 2, channel 3, nominal current	USINT	1	0	Read
26	Module 2, channel 3, currently flowing current	USINT	1	0	Read
27	Module 2, channel 4, status byte	USINT	1	0	Read
28	Module 2, channel 4, nominal current	USINT	1	0	Read
29	Module 2, channel 4, currently flowing current	USINT	1	0	Read
30	Module 3, channel 1, status byte	USINT	1	0	Read
31	Module 3, channel 1, nominal current	USINT	1	0	Read
32	Module 3, channel 1, currently flowing current	USINT	1	0	Read
33	Module 3, channel 2, status byte	USINT	1	0	Read

Table 7-4 Structure of the input assembly [...]

Byte offset	Description	EIP data type	Data size	Default	Read/write
34	Module 3, channel 2, nominal current	USINT	1	0	Read
35	Module 3, channel 2, currently flowing current	USINT	1	0	Read
36	Module 3, channel 3, status byte	USINT	1	0	Read
37	Module 3, channel 3, nominal current	USINT	1	0	Read
38	Module 3, channel 3, currently flowing current	USINT	1	0	Read
39	Module 3, channel 4, status byte	USINT	1	0	Read
40	Module 3, channel 4, nominal current	USINT	1	0	Read
41	Module 3, channel 4, currently flowing current	USINT	1	0	Read
42	Module 4, channel 1, status byte	USINT	1	0	Read
43	Module 4, channel 1, nominal current	USINT	1	0	Read
44	Module 4, channel 1, currently flowing current	USINT	1	0	Read
45	Module 4, channel 2, status byte	USINT	1	0	Read
46	Module 4, channel 2, nominal current	USINT	1	0	Read
47	Module 4, channel 2, currently flowing current	USINT	1	0	Read
48	Module 4, channel 3, status byte	USINT	1	0	Read
49	Module 4, channel 3, nominal current	USINT	1	0	Read
50	Module 4, channel 3, currently flowing current	USINT	1	0	Read
51	Module 4, channel 4, status byte	USINT	1	0	Read
52	Module 4, channel 4, nominal current	USINT	1	0	Read
53	Module 4, channel 4, currently flowing current	USINT	1	0	Read
54	Module 5, channel 1, status byte	USINT	1	0	Read
55	Module 5, channel 1, nominal current	USINT	1	0	Read
56	Module 5, channel 1, currently flowing current	USINT	1	0	Read
57	Module 5, channel 2, status byte	USINT	1	0	Read
58	Module 5, channel 2, nominal current	USINT	1	0	Read
59	Module 5, channel 2, currently flowing current	USINT	1	0	Read
60	Module 5, channel 3, status byte	USINT	1	0	Read
61	Module 5, channel 3, nominal current	USINT	1	0	Read
62	Module 5, channel 3, currently flowing current	USINT	1	0	Read
63	Module 5, channel 4, status byte	USINT	1	0	Read
64	Module 5, channel 4, nominal current	USINT	1	0	Read
65	Module 5, channel 4, currently flowing current	USINT	1	0	Read
66	Module 6, channel 1, status byte	USINT	1	0	Read
67	Module 6, channel 1, nominal current	USINT	1	0	Read
68	Module 6, channel 1, currently flowing current	USINT	1	0	Read
69	Module 6, channel 2, status byte	USINT	1	0	Read
70	Module 6, channel 2, nominal current	USINT	1	0	Read

Table 7-4 Structure of the input assembly [...]

Byte offset	Description	EIP data type	Data size	Default	Read/write
71	Module 6, channel 2, currently flowing current	USINT	1	0	Read
72	Module 6, channel 3, status byte	USINT	1	0	Read
73	Module 6, channel 3, nominal current	USINT	1	0	Read
74	Module 6, channel 3, currently flowing current	USINT	1	0	Read
75	Module 6, channel 4, status byte	USINT	1	0	Read
76	Module 6, channel 4, nominal current	USINT	1	0	Read
77	Module 6, channel 4, currently flowing current	USINT	1	0	Read
78	Module 7, channel 1, status byte	USINT	1	0	Read
79	Module 7, channel 1, nominal current	USINT	1	0	Read
80	Module 7, channel 1, currently flowing current	USINT	1	0	Read
81	Module 7, channel 2, status byte	USINT	1	0	Read
82	Module 7, channel 2, nominal current	USINT	1	0	Read
83	Module 7, channel 2, currently flowing current	USINT	1	0	Read
84	Module 7, channel 3, status byte	USINT	1	0	Read
85	Module 7, channel 3, nominal current	USINT	1	0	Read
86	Module 7, channel 3, currently flowing current	USINT	1	0	Read
87	Module 7, channel 4, status byte	USINT	1	0	Read
88	Module 7, channel 4, nominal current	USINT	1	0	Read
89	Module 7, channel 4, currently flowing current	USINT	1	0	Read
90	Module 8, channel 1, status byte	USINT	1	0	Read
91	Module 8, channel 1, nominal current	USINT	1	0	Read
92	Module 8, channel 1, currently flowing current	USINT	1	0	Read
93	Module 8, channel 2, status byte	USINT	1	0	Read
94	Module 8, channel 2, nominal current	USINT	1	0	Read
95	Module 8, channel 2, currently flowing current	USINT	1	0	Read
96	Module 8, channel 3, status byte	USINT	1	0	Read
97	Module 8, channel 3, nominal current	USINT	1	0	Read
98	Module 8, channel 3, currently flowing current	USINT	1	0	Read
99	Module 8, channel 4, status byte	USINT	1	0	Read
100	Module 8, channel 4, nominal current	USINT	1	0	Read
101	Module 8, channel 4, currently flowing current	USINT	1	0	Read
102	Module 9, channel 1, status byte	USINT	1	0	Read
103	Module 9, channel 1, nominal current	USINT	1	0	Read
104	Module 9, channel 1, currently flowing current	USINT	1	0	Read
105	Module 9, channel 2, status byte	USINT	1	0	Read
106	Module 9, channel 2, nominal current	USINT	1	0	Read
107	Module 9, channel 2, currently flowing current	USINT	1	0	Read

Table 7-4 Structure of the input assembly [...]

Byte offset	Description	EIP data type	Data size	Default	Read/write
108	Module 9, channel 3, status byte	USINT	1	0	Read
109	Module 9, channel 3, nominal current	USINT	1	0	Read
110	Module 9, channel 3, currently flowing current	USINT	1	0	Read
111	Module 9, channel 4, status byte	USINT	1	0	Read
112	Module 9, channel 4, nominal current	USINT	1	0	Read
113	Module 9, channel 4, currently flowing current	USINT	1	0	Read
114	Module 10, channel 1, status byte	USINT	1	0	Read
115	Module 10, channel 1, nominal current	USINT	1	0	Read
116	Module 10, channel 1, currently flowing current	USINT	1	0	Read
117	Module 10, channel 2, status byte	USINT	1	0	Read
118	Module 10, channel 2, nominal current	USINT	1	0	Read
119	Module 10, channel 2, currently flowing current	USINT	1	0	Read
120	Module 10, channel 3, status byte	USINT	1	0	Read
121	Module 10, channel 3, nominal current	USINT	1	0	Read
122	Module 10, channel 3, currently flowing current	USINT	1	0	Read
123	Module 10, channel 4, status byte	USINT	1	0	Read
124	Module 10, channel 4, nominal current	USINT	1	0	Read
125	Module 10, channel 4, currently flowing current	USINT	1	0	Read
126	Module 11, channel 1, status byte	USINT	1	0	Read
127	Module 11, channel 1, nominal current	USINT	1	0	Read
128	Module 11, channel 1, currently flowing current	USINT	1	0	Read
129	Module 11, channel 2, status byte	USINT	1	0	Read
130	Module 11, channel 2, nominal current	USINT	1	0	Read
131	Module 11, channel 2, currently flowing current	USINT	1	0	Read
132	Module 11, channel 3, status byte	USINT	1	0	Read
133	Module 11, channel 3, nominal current	USINT	1	0	Read
134	Module 11, channel 3, currently flowing current	USINT	1	0	Read
135	Module 11, channel 4, status byte	USINT	1	0	Read
136	Module 11, channel 4, nominal current	USINT	1	0	Read
137	Module 11, channel 4, currently flowing current	USINT	1	0	Read
138	Module 12, channel 1, status byte	USINT	1	0	Read
139	Module 12, channel 1, nominal current	USINT	1	0	Read
140	Module 12, channel 1, currently flowing current	USINT	1	0	Read
141	Module 12, channel 2, status byte	USINT	1	0	Read
142	Module 12, channel 2, nominal current	USINT	1	0	Read
143	Module 12, channel 2, currently flowing current	USINT	1	0	Read
144	Module 12, channel 3, status byte	USINT	1	0	Read

Table 7-4 Structure of the input assembly [...]

Byte offset	Description	EIP data type	Data size	Default	Read/write
145	Module 12, channel 3, nominal current	USINT	1	0	Read
146	Module 12, channel 3, currently flowing current	USINT	1	0	Read
147	Module 12, channel 4, status byte	USINT	1	0	Read
148	Module 12, channel 4, nominal current	USINT	1	0	Read
149	Module 12, channel 4, currently flowing current	USINT	1	0	Read
150	Module 13, channel 1, status byte	USINT	1	0	Read
151	Module 13, channel 1, nominal current	USINT	1	0	Read
152	Module 13, channel 1, currently flowing current	USINT	1	0	Read
153	Module 13, channel 2, status byte	USINT	1	0	Read
154	Module 13, channel 2, nominal current	USINT	1	0	Read
155	Module 13, channel 2, currently flowing current	USINT	1	0	Read
156	Module 13, channel 3, status byte	USINT	1	0	Read
157	Module 13, channel 3, nominal current	USINT	1	0	Read
158	Module 13, channel 3, currently flowing current	USINT	1	0	Read
159	Module 13, channel 4, status byte	USINT	1	0	Read
160	Module 13, channel 4, nominal current	USINT	1	0	Read
161	Module 13, channel 4, currently flowing current	USINT	1	0	Read
162	Module 14, channel 1, status byte	USINT	1	0	Read
163	Module 14, channel 1, nominal current	USINT	1	0	Read
164	Module 14, channel 1, currently flowing current	USINT	1	0	Read
165	Module 14, channel 2, status byte	USINT	1	0	Read
166	Module 14, channel 2, nominal current	USINT	1	0	Read
167	Module 14, channel 2, currently flowing current	USINT	1	0	Read
168	Module 14, channel 3, status byte	USINT	1	0	Read
169	Module 14, channel 3, nominal current	USINT	1	0	Read
170	Module 14, channel 3, currently flowing current	USINT	1	0	Read
171	Module 14, channel 4, status byte	USINT	1	0	Read
172	Module 14, channel 4, nominal current	USINT	1	0	Read
173	Module 14, channel 4, currently flowing current	USINT	1	0	Read
174	Module 15, channel 1, status byte	USINT	1	0	Read
175	Module 15, channel 1, nominal current	USINT	1	0	Read
176	Module 15, channel 1, currently flowing current	USINT	1	0	Read
177	Module 15, channel 2, status byte	USINT	1	0	Read
178	Module 15, channel 2, nominal current	USINT	1	0	Read
179	Module 15, channel 2, currently flowing current	USINT	1	0	Read
180	Module 15, channel 3, status byte	USINT	1	0	Read
181	Module 15, channel 3, nominal current	USINT	1	0	Read

Table 7-4 Structure of the input assembly [...]

Byte offset	Description	EIP data type	Data size	Default	Read/write
182	Module 15, channel 3, currently flowing current	USINT	1	0	Read
183	Module 15, channel 4, status byte	USINT	1	0	Read
184	Module 15, channel 4, nominal current	USINT	1	0	Read
185	Module 15, channel 4, currently flowing current	USINT	1	0	Read
186	Module 16, channel 1, status byte	USINT	1	0	Read
187	Module 16, channel 1, nominal current	USINT	1	0	Read
188	Module 16, channel 1, currently flowing current	USINT	1	0	Read
189	Module 16, channel 2, status byte	USINT	1	0	Read
190	Module 16, channel 2, nominal current	USINT	1	0	Read
191	Module 16, channel 2, currently flowing current	USINT	1	0	Read
192	Module 16, channel 3, status byte	USINT	1	0	Read
193	Module 16, channel 3, nominal current	USINT	1	0	Read
194	Module 16, channel 3, currently flowing current	USINT	1	0	Read
195	Module 16, channel 4, status byte	USINT	1	0	Read
196	Module 16, channel 4, nominal current	USINT	1	0	Read
197	Module 16, channel 4, currently flowing current	USINT	1	0	Read
198 ... 199	QUINT POWER status register	INT	2	0	Read
200	Reserved	INT	2	0	Read
202	Reserved	INT	2	0	Read
204	Reserved	INT	2	0	Read
206	Reserved	INT	2	0	Read

7.2.1 Input assembly, global status

Byte 0 in the input assembly represents the current system status and is structured as follows:

Table 7-5 Input assembly, global status

Global status	Bit	Function	Value range
Byte 0	0	Undervoltage	0 = Voltage OK
			1 = Voltage below 9.2 V
	1	Overvoltage	0 = Voltage OK
			1 = Voltage above 30.5 V
	2	System error (group message)	0 = No errors
			1 = At least one error in the system
	3	80% nominal current warning	0 = No 80% nominal current warning
			1 = At least one 80% nominal current warning in the system
	4	Total current shutdown	0 = No total current shutdown
			1 = At least one module switched off due to excessively high total system current
	5	Reserved	
	6	Reserved	
	7	Processing of the config assembly	0 = Processing completed
			1 = Processing in progress

7.2.2 Input assembly, global module counter

Byte 1 in the input assembly indicates the number of circuit breaker modules installed in the system and is structured as follows:

Table 7-6 Input assembly, global module counter

Global module counter	Bit	Function	Value range
Byte 0	0 ... 7	Number of circuit breaker modules installed in the system	0 ... 16

7.2.3 Input assembly, global total current

Byte 2 in the input assembly indicates the total current that flows via the CAPAROC system.

The information can be retrieved as follows:

Table 7-7 Input assembly, global total current

Global total current	Bit	Function	Value range
Byte 0	0 ... 7	Total current (low byte)	0 ... 500 (0 A to 50.0 A)
Byte 1	8 ... 15	Total current (high byte)	

7.2.4 Input assembly, global input voltage

Byte 3 of the input assembly contains information on the input voltage of the CAPAROC system.

Table 7-8 Input process data CAPAROC E4 12-24DC/1-10A

Global input voltage	Bit	Function	Value range
Byte 0	0 ... 7	Total voltage (low byte)	900 ... 3050 (9.0 V to 30.5 V)
Byte 1	8 ... 15	Total voltage (high byte)	

7.2.5 Input assembly, module data block

Every circuit breaker module connected to the power module has a data block with 12 bytes in the input assembly.

That is equivalent to 3 bytes per channel for 4-channel modules. Accordingly, the bytes are not filled with information for 1-channel or 2-channel modules.

Table 7-9 Output process data CAPAROC E4 12-24DC/1-10A

Module data block	Bit	Function	Value range
Byte 0	0	Channel 1, status	0 = Channel switched off
			1 = Channel switched on
	1	Channel 1, 80% warning	0 = Flowing current less than 80% of nominal current
			1 = Flowing current greater than/equal to 80% of nominal current
	2	Channel 1, overload tripping	0 = No overload tripping
			1 = Channel tripped due to overload
	3	Channel 1, short-circuit tripping	0 = No short-circuit tripping
			1 = Channel tripped due to short circuit
	4	Channel 1, hardware fault	0 = Hardware OK
			1 = Hardware fault
	5	Channel 1, total current shutdown	0 = No total current shutdown
			1 = Channel switched off due to excessively high total current
	6 ... 7	Reserve	
Byte 1	0 ... 7	Nominal current	1 A ... 10 A
Byte 2	0 ... 7	Flowing current	0 ... 255 (0 A to 25.5 A)
Byte 3	0	Channel 2, status	0 = Channel switched off
			1 = Channel switched on
	1	Channel 2, 80% warning	0 = Flowing current less than 80% of nominal current
			1 = Flowing current greater than/equal to 80% of nominal current
	2	Channel 2, overload tripping	0 = No overload tripping
			1 = Channel tripped due to overload
	3	Channel 2, short-circuit tripping	0 = No short-circuit tripping
			1 = Channel tripped due to short circuit
	4	Channel 2, hardware fault	0 = Hardware OK
			1 = Hardware fault
	5	Channel 2, total current shutdown	0 = No total current shutdown
			1 = Channel switched off due to excessively high total current
	6 ... 7	Reserved	
Byte 4	0 ... 7	Nominal current	1 A ... 10 A
Byte 5	0 ... 7	Flowing current	0 ... 255 (0 A to 25.5 A)

Table 7-9 Output process data CAPAROC E4 12-24DC/1-10A [...]

Module data block	Bit	Function	Value range
Byte 6	0	Channel 3, status	0 = Channel switched off
			1 = Channel switched on
	1	Channel 3, 80% warning	0 = Flowing current less than 80% of nominal current
			1 = Flowing current greater than/equal to 80% of nominal current
	2	Channel 3, overload tripping	0 = No overload tripping
			1 = Channel tripped due to overload
	3	Channel 3, short-circuit tripping	0 = No short-circuit tripping
			1 = Channel tripped due to short circuit
	4	Channel 3, hardware fault	0 = Hardware OK
			1 = Hardware fault
	5	Channel 3, total current shutdown	0 = No total current shutdown
			1 = Channel switched off due to excessively high total current
	6 ... 7	Reserved	
Byte 7	0 ... 7	Nominal current	1 A ... 10 A
Byte 8	0 ... 7	Flowing current	0 ... 255 (0 A to 25.5 A)
Byte 9	0	Channel 4, status	0 = Channel switched off
			1 = Channel switched on
	1	Channel 4, 80% warning	0 = Flowing current less than 80% of nominal current
			1 = Flowing current greater than/equal to 80% of nominal current
	2	Channel 4, overload tripping	0 = No overload tripping
			1 = Channel tripped due to overload
	3	Channel 4, short-circuit tripping	0 = No short-circuit tripping
			1 = Channel tripped due to short circuit
	4	Channel 4, hardware fault	0 = Hardware OK
			1 = Hardware fault
	5	Channel 4, total current shutdown	0 = No total current shutdown
			1 = Channel switched off due to excessively high total current
	6 ... 7	Reserved	
Byte 10	0 ... 7	Nominal current	1 A ... 10 A
Byte 11	0 ... 7	Flowing current	0 ... 255 (0 A to 25.5 A)

7.2.6 Input assembly, QUINT POWER status register

Data of the connected power supply can be retrieved via bytes 198 and 199.

The prerequisite for this is a Quint power supply that is connected to the system bus interface of the CAPAROC PM module.

The precise function of the various parameters can be found in the manual for the corresponding power supply.

Table 7-10 Input assembly, QUINT POWER status register

QUINT POWER status register	Bit	Function	Value range
Byte 0	0	DC OK	0 = DC voltage OK
			1 = DC voltage below the warning threshold
	1	Output current status	0 = Output current OK
			1 = Output current above the warning threshold
	2	Status of operating hours	0 = Operating hours below the warning threshold
			1 = Operating hours above the warning threshold
	3	Temperature warning status	0 = No warning
			1 = Temperature warning
	4	Reserved	
	5	Voltage limitation status	0 = Voltage limitation not active
			1 = Voltage limitation active
	6	Phase monitoring status	0 = Phases OK (3AC)
			1 = Phase error (2AC)
	7	Status of service operating hours	0 = Remaining operating hours above the warning threshold
			1 = Remaining operating hours below the warning threshold

Table 7-10 Input assembly, [...]QUINT POWER status register

QUINT POWER status register	Bit	Function	Value range
Byte 1	8 ... 10	Operating mode	0 = Normal mode
			1 = Startup
			2 = No AC input
			3 = Restart
			4 = Fuse mode
			5 = Sleep mode
			6 = Low AC input voltage
			7 = Smart HICCUP operation
	11 ... 12	Direction of rotating field	0 = Right
			1 = Left
			2 = No information (e.g., DC or 2-phase operation)
	13	AC/DC detection of input voltage	0 = AC
			1 = DC voltage below the warning threshold
	14 ... 15	Reserved	

7.3 Structure of the config assembly

In the event of incorrect programming of the config assembly, the first error detected is output via the extended/additional error code.

The faulty parameter can be detected due to the message.

Table 7-11 Structure of the config assembly

EDS parame-ter no.	Description	Data type	Data size	Min. value	Max. value	Default setting	Read/write
1	Global nominal current lock	USINT	1	0	2	2	Read and write
2	Global user interface lock	USINT	1	0	2	2	Read and write
3	Global switch-on delay	INT	2	0	10000	10000	Read and write
4	Global operating mode	USINT	1	0	2	2	Read and write
5	Reserved	USINT	1	0	255	2	Read and write
6	Module 1, channel 1, nominal current	USINT	1	0	20	0	Read and write
7	Module 1, channel 1, programming lock	USINT	1	0	2	2	Read and write
8	Module 1, channel 1, status	USINT	1	0	2	2	Read and write
9	Module 1, channel 2, nominal current	USINT	1	0	20	0	Read and write
10	Module 1, channel 2, programming lock	USINT	1	0	2	2	Read and write
11	Module 1, channel 2, status	USINT	1	0	2	2	Read and write
12	Module 1, channel 3, nominal current	USINT	1	0	20	0	Read and write
13	Module 1, channel 3, programming lock	USINT	1	0	2	2	Read and write
14	Module 1, channel 3, status	USINT	1	0	2	2	Read and write
15	Module 1, channel 4, nominal current	USINT	1	0	20	0	Read and write
16	Module 1, channel 4, programming lock	USINT	1	0	2	2	Read and write
17	Module 1, channel 4, status	USINT	1	0	2	2	Read and write
18	Module 2, channel 1, nominal current	USINT	1	0	20	0	Read and write
19	Module 2, channel 1, programming lock	USINT	1	0	2	2	Read and write
20	Module 2, channel 1, status	USINT	1	0	2	2	Read and write
21	Module 2, channel 2, nominal current	USINT	1	0	20	0	Read and write
22	Module 2, channel 2, programming lock	USINT	1	0	2	2	Read and write
23	Module 2, channel 2, status	USINT	1	0	2	2	Read and write
24	Module 2, channel 3, nominal current	USINT	1	0	20	0	Read and write
25	Module 2, channel 3, programming lock	USINT	1	0	2	2	Read and write
26	Module 2, channel 3, status	USINT	1	0	2	2	Read and write
27	Module 2, channel 4, nominal current	USINT	1	0	20	0	Read and write
28	Module 2, channel 4, programming lock	USINT	1	0	2	2	Read and write
29	Module 2, channel 4, status	USINT	1	0	2	2	Read and write
30	Module 3, channel 1, nominal current	USINT	1	0	20	0	Read and write
31	Module 3, channel 1, programming lock	USINT	1	0	2	2	Read and write

Table 7-11 [...]Structure of the config assembly

EDS parame-ter no.	Description	Data type	Data size	Min. valu e	Max. value	Default setting	Read/write
32	Module 3, channel 1, status	USINT	1	0	2	2	Read and write
33	Module 3, channel 2, nominal current	USINT	1	0	20	0	Read and write
34	Module 3, channel 2, programming lock	USINT	1	0	2	2	Read and write
35	Module 3, channel 2, status	USINT	1	0	2	2	Read and write
36	Module 3, channel 3, nominal current	USINT	1	0	20	0	Read and write
37	Module 3, channel 3, programming lock	USINT	1	0	2	2	Read and write
38	Module 3, channel 3, status	USINT	1	0	2	2	Read and write
39	Module 3, channel 4, nominal current	USINT	1	0	20	0	Read and write
40	Module 3, channel 4, programming lock	USINT	1	0	2	2	Read and write
41	Module 3, channel 4, status	USINT	1	0	2	2	Read and write
42	Module 4, channel 1, nominal current	USINT	1	0	20	0	Read and write
43	Module 4, channel 1, programming lock	USINT	1	0	2	2	Read and write
44	Module 4, channel 1, status	USINT	1	0	2	2	Read and write
45	Module 4, channel 2, nominal current	USINT	1	0	20	0	Read and write
46	Module 4, channel 2, programming lock	USINT	1	0	2	2	Read and write
47	Module 4, channel 2, status	USINT	1	0	2	2	Read and write
48	Module 4, channel 3, nominal current	USINT	1	0	20	0	Read and write
49	Module 4, channel 3, programming lock	USINT	1	0	2	2	Read and write
50	Module 4, channel 3, status	USINT	1	0	2	2	Read and write
51	Module 4, channel 4, nominal current	USINT	1	0	20	0	Read and write
52	Module 4, channel 4, programming lock	USINT	1	0	2	2	Read and write
53	Module 4, channel 4, status	USINT	1	0	2	2	Read and write
54	Module 5, channel 1, nominal current	USINT	1	0	20	0	Read and write
55	Module 5, channel 1, programming lock	USINT	1	0	2	2	Read and write
56	Module 5, channel 1, status	USINT	1	0	2	2	Read and write
57	Module 5, channel 2, nominal current	USINT	1	0	20	0	Read and write
58	Module 5, channel 2, programming lock	USINT	1	0	2	2	Read and write
59	Module 5, channel 2, status	USINT	1	0	2	2	Read and write
60	Module 5, channel 3, nominal current	USINT	1	0	20	0	Read and write
61	Module 5, channel 3, programming lock	USINT	1	0	2	2	Read and write
62	Module 5, channel 3, status	USINT	1	0	2	2	Read and write
63	Module 5, channel 4, nominal current	USINT	1	0	20	0	Read and write
64	Module 5, channel 4, programming lock	USINT	1	0	2	2	Read and write
65	Module 5, channel 4, status	USINT	1	0	2	2	Read and write
66	Module 6, channel 1, nominal current	USINT	1	0	20	0	Read and write
67	Module 6, channel 1, programming lock	USINT	1	0	2	2	Read and write

Table 7-11 [...]Structure of the config assembly

EDS parame-ter no.	Description	Data type	Data size	Min. valu e	Max. value	Default setting	Read/write
68	Module 6, channel 1, status	USINT	1	0	2	2	Read and write
69	Module 6, channel 2, nominal current	USINT	1	0	20	0	Read and write
70	Module 6, channel 2, programming lock	USINT	1	0	2	2	Read and write
71	Module 6, channel 2, status	USINT	1	0	2	2	Read and write
72	Module 6, channel 3, nominal current	USINT	1	0	20	0	Read and write
73	Module 6, channel 3, programming lock	USINT	1	0	2	2	Read and write
74	Module 6, channel 3, status	USINT	1	0	2	2	Read and write
75	Module 6, channel 4, nominal current	USINT	1	0	20	0	Read and write
76	Module 6, channel 4, programming lock	USINT	1	0	2	2	Read and write
77	Module 6, channel 4, status	USINT	1	0	2	2	Read and write
78	Module 7, channel 1, nominal current	USINT	1	0	20	0	Read and write
79	Module 7, channel 1, programming lock	USINT	1	0	2	2	Read and write
80	Module 7, channel 1, status	USINT	1	0	2	2	Read and write
81	Module 7, channel 2, nominal current	USINT	1	0	20	0	Read and write
82	Module 7, channel 2, programming lock	USINT	1	0	2	2	Read and write
83	Module 7, channel 2, status	USINT	1	0	2	2	Read and write
84	Module 7, channel 3, nominal current	USINT	1	0	20	0	Read and write
85	Module 7, channel 3, programming lock	USINT	1	0	2	2	Read and write
86	Module 7, channel 3, status	USINT	1	0	2	2	Read and write
87	Module 7, channel 4, nominal current	USINT	1	0	20	0	Read and write
88	Module 7, channel 4, programming lock	USINT	1	0	2	2	Read and write
89	Module 7, channel 4, status	USINT	1	0	2	2	Read and write
90	Module 8, channel 1, nominal current	USINT	1	0	20	0	Read and write
91	Module 8, channel 1, programming lock	USINT	1	0	2	2	Read and write
92	Module 8, channel 1, status	USINT	1	0	2	2	Read and write
93	Module 8, channel 2, nominal current	USINT	1	0	20	0	Read and write
94	Module 8, channel 2, programming lock	USINT	1	0	2	2	Read and write
95	Module 8, channel 2, status	USINT	1	0	2	2	Read and write
96	Module 8, channel 3, nominal current	USINT	1	0	20	0	Read and write
97	Module 8, channel 3, programming lock	USINT	1	0	2	2	Read and write
98	Module 8, channel 3, status	USINT	1	0	2	2	Read and write
99	Module 8, channel 4, nominal current	USINT	1	0	20	0	Read and write
100	Module 8, channel 4, programming lock	USINT	1	0	2	2	Read and write
101	Module 8, channel 4, status	USINT	1	0	2	2	Read and write
102	Module 9, channel 1, nominal current	USINT	1	0	20	0	Read and write
103	Module 9, channel 1, programming lock	USINT	1	0	2	2	Read and write

Table 7-11 [...]Structure of the config assembly

EDS parame-ter no.	Description	Data type	Data size	Min. valu e	Max. value	Default setting	Read/write
104	Module 9, channel 1, status	USINT	1	0	2	2	Read and write
105	Module 9, channel 2, nominal current	USINT	1	0	20	0	Read and write
106	Module 9, channel 2, programming lock	USINT	1	0	2	2	Read and write
107	Module 9, channel 2, status	USINT	1	0	2	2	Read and write
108	Module 9, channel 3, nominal current	USINT	1	0	20	0	Read and write
109	Module 9, channel 3, programming lock	USINT	1	0	2	2	Read and write
110	Module 9, channel 3, status	USINT	1	0	2	2	Read and write
111	Module 9, channel 4, nominal current	USINT	1	0	20	0	Read and write
112	Module 9, channel 4, programming lock	USINT	1	0	2	2	Read and write
113	Module 9, channel 4, status	USINT	1	0	2	2	Read and write
114	Module 10, channel 1, nominal current	USINT	1	0	20	0	Read and write
115	Module 10, channel 1, programming lock	USINT	1	0	2	2	Read and write
116	Module 10, channel 1, status	USINT	1	0	2	2	Read and write
117	Module 10, channel 2, nominal current	USINT	1	0	20	0	Read and write
118	Module 10, channel 2, programming lock	USINT	1	0	2	2	Read and write
119	Module 10, channel 2, status	USINT	1	0	2	2	Read and write
120	Module 10, channel 3, nominal current	USINT	1	0	20	0	Read and write
121	Module 10, channel 3, programming lock	USINT	1	0	2	2	Read and write
122	Module 10, channel 3, status	USINT	1	0	2	2	Read and write
123	Module 10, channel 4, nominal current	USINT	1	0	20	0	Read and write
124	Module 10, channel 4, programming lock	USINT	1	0	2	2	Read and write
125	Module 10, channel 4, status	USINT	1	0	2	2	Read and write
126	Module 11, channel 1, nominal current	USINT	1	0	20	0	Read and write
127	Module 11, channel 1, programming lock	USINT	1	0	2	2	Read and write
128	Module 11, channel 1, status	USINT	1	0	2	2	Read and write
129	Module 11, channel 2, nominal current	USINT	1	0	20	0	Read and write
130	Module 11, channel 2, programming lock	USINT	1	0	2	2	Read and write
131	Module 11, channel 2, status	USINT	1	0	2	2	Read and write
132	Module 11, channel 3, nominal current	USINT	1	0	20	0	Read and write
133	Module 11, channel 3, programming lock	USINT	1	0	2	2	Read and write
134	Module 11, channel 3, status	USINT	1	0	2	2	Read and write
135	Module 11, channel 4, nominal current	USINT	1	0	20	0	Read and write
136	Module 11, channel 4, programming lock	USINT	1	0	2	2	Read and write
137	Module 11, channel 4, status	USINT	1	0	2	2	Read and write
138	Module 12, channel 1, nominal current	USINT	1	0	20	0	Read and write
139	Module 12, channel 1, programming lock	USINT	1	0	2	2	Read and write

Table 7-11 [...]Structure of the config assembly

EDS parame-ter no.	Description	Data type	Data size	Min. valu e	Max. value	Default setting	Read/write
140	Module 12, channel 1, status	USINT	1	0	2	2	Read and write
141	Module 12, channel 2, nominal current	USINT	1	0	20	0	Read and write
142	Module 12, channel 2, programming lock	USINT	1	0	2	2	Read and write
143	Module 12, channel 2, status	USINT	1	0	2	2	Read and write
144	Module 12, channel 3, nominal current	USINT	1	0	20	0	Read and write
145	Module 12, channel 3, programming lock	USINT	1	0	2	2	Read and write
146	Module 12, channel 3, status	USINT	1	0	2	2	Read and write
147	Module 12, channel 4, nominal current	USINT	1	0	20	0	Read and write
148	Module 12, channel 4, programming lock	USINT	1	0	2	2	Read and write
149	Module 12, channel 4, status	USINT	1	0	2	2	Read and write
150	Module 13, channel 1, nominal current	USINT	1	0	20	0	Read and write
151	Module 13, channel 1, programming lock	USINT	1	0	2	2	Read and write
152	Module 13, channel 1, status	USINT	1	0	2	2	Read and write
153	Module 13, channel 2, nominal current	USINT	1	0	20	0	Read and write
154	Module 13, channel 2, programming lock	USINT	1	0	2	2	Read and write
155	Module 13, channel 2, status	USINT	1	0	2	2	Read and write
156	Module 13, channel 3, nominal current	USINT	1	0	20	0	Read and write
157	Module 13, channel 3, programming lock	USINT	1	0	2	2	Read and write
158	Module 13, channel 3, status	USINT	1	0	2	2	Read and write
159	Module 13, channel 4, nominal current	USINT	1	0	20	0	Read and write
160	Module 13, channel 4, programming lock	USINT	1	0	2	2	Read and write
161	Module 13, channel 4, status	USINT	1	0	2	2	Read and write
162	Module 14, channel 1, nominal current	USINT	1	0	20	0	Read and write
163	Module 14, channel 1, programming lock	USINT	1	0	2	2	Read and write
164	Module 14, channel 1, status	USINT	1	0	2	2	Read and write
165	Module 14, channel 2, nominal current	USINT	1	0	20	0	Read and write
166	Module 14, channel 2, programming lock	USINT	1	0	2	2	Read and write
167	Module 14, channel 2, status	USINT	1	0	2	2	Read and write
168	Module 14, channel 3, nominal current	USINT	1	0	20	0	Read and write
169	Module 14, channel 3, programming lock	USINT	1	0	2	2	Read and write
170	Module 14, channel 3, status	USINT	1	0	2	2	Read and write
171	Module 14, channel 4, nominal current	USINT	1	0	20	0	Read and write
172	Module 14, channel 4, programming lock	USINT	1	0	2	2	Read and write
173	Module 14, channel 4, status	USINT	1	0	2	2	Read and write
174	Module 15, channel 1, nominal current	USINT	1	0	20	0	Read and write
175	Module 15, channel 1, programming lock	USINT	1	0	2	2	Read and write

Table 7-11 [...]Structure of the config assembly

EDS parame-ter no.	Description	Data type	Data size	Min. valu e	Max. value	Default setting	Read/write
176	Module 15, channel 1, status	USINT	1	0	2	2	Read and write
177	Module 15, channel 2, nominal current	USINT	1	0	20	0	Read and write
178	Module 15, channel 2, programming lock	USINT	1	0	2	2	Read and write
179	Module 15, channel 2, status	USINT	1	0	2	2	Read and write
180	Module 15, channel 3, nominal current	USINT	1	0	20	0	Read and write
181	Module 15, channel 3, programming lock	USINT	1	0	2	2	Read and write
182	Module 15, channel 3, status	USINT	1	0	2	2	Read and write
183	Module 15, channel 4, nominal current	USINT	1	0	20	0	Read and write
184	Module 15, channel 4, programming lock	USINT	1	0	2	2	Read and write
185	Module 15, channel 4, status	USINT	1	0	2	2	Read and write
186	Module 16, channel 1, nominal current	USINT	1	0	20	0	Read and write
187	Module 16, channel 1, programming lock	USINT	1	0	2	2	Read and write
188	Module 16, channel 1, status	USINT	1	0	2	2	Read and write
189	Module 16, channel 2, nominal current	USINT	1	0	20	0	Read and write
190	Module 16, channel 2, programming lock	USINT	1	0	2	2	Read and write
191	Module 16, channel 2, status	USINT	1	0	2	2	Read and write
192	Module 16, channel 3, nominal current	USINT	1	0	20	0	Read and write
193	Module 16, channel 3, programming lock	USINT	1	0	2	2	Read and write
194	Module 16, channel 3, status	USINT	1	0	2	2	Read and write
195	Module 16, channel 4, nominal current	USINT	1	0	20	0	Read and write
196	Module 16, channel 4, programming lock	USINT	1	0	2	2	Read and write
197	Module 16, channel 4, status	USINT	1	0	2	2	Read and write
198	Reserved	INT	2	0	0	0	
199	QUINT POWER, output voltage	DINT	4	2390	2960	2410	Write
200	QUINT POWER Operating mode	DINT	4	0	7	0	Read and write
201	QUINT POWER, relay contact	DINT	4	0	65535	0	Read and write
202	QUINT POWER, output voltage limit	DINT	4	25	135	90	Read and write
203	QUINT POWER, output current limit	DINT	4	5	200	100	Read and write
204	QUINT POWER, limit for operating hours	DINT	4	0	32768	3650	Read and write
205	QUINT POWER, limit for remaining oper-ating hours	DINT	4	0	5475	0	Read and write
206	Reserved	DINT	4	0	0	0	
207	Reserved	DINT	4	0	0	0	
208	Reserved	DINT	4	0	0	0	
209	Reserved	DINT	4	0	0	0	

7.3.1 Config assembly, global nominal current lock

Using the parameter of the global nominal current lock, all channels can be locked centrally on the hardware side for a change of the nominal currents at the circuit breakers.

Table 7-12 Config assembly, global nominal current lock

Global nominal current lock	Data type	Bit	Function	Default	Read/write
Byte 0	USINT	0 ... 7	0 = Lock inactive	0	Read/write
			1 = Lock active		
			2 = No change (EDS standard)		

7.3.2 Config assembly, global user interface lock

The global user interface lock is used to completely lock all circuit breakers on the hardware side.

Control via parameter class (0x0F) or the config assembly is still possible.

Table 7-13 Config assembly, global user interface lock

Global user interface lock	Data type	Bit	Function	Default	Read/write
Byte 0	USINT	0 ... 7	0 = Lock inactive	0	Read/write
			1 = Lock active		
			2 = No change (EDS standard)		

7.3.3 Config assembly, global switch-on delay

This parameter defines the switch-on delay between two channels in the CAPAROC system. This means that cascading switching on is possible, thus reducing the power supply load.

Table 7-14 Config assembly, global switch-on delay

Global switch-on delay	Data type	Bit	Function	Default	Read/write
Byte 0	INT	0 ... 7	0 = Min. delay	0	Read/write
			25 = 25 ms		
			50 = 50 ms		
			100 = 100 ms		
Byte 1		8 ... 15	150 = 150 ms		
			200 = 200 ms		
			500 = 500 ms		
			10000 = No change (EDS standard)		

Due to internal system processes, there is always a minimal delay. Therefore, the value 0 does not always stand for 0 ms.

7.3.4 Config assembly, global operating mode

Global operating mode allows the CAPAROC system to be configured to wait for the fieldbus connection before it loads the startup parameters.

Table 7-15 Config assembly, global operating mode

Global operating mode	Data type	Bit	Function	Default	Read/write
Byte 0	USINT	0 ... 7	0 = Independent operation	0	Read/write
			1 = Wait for fieldbus		
			2 = No change (EDS standard)		

7.3.5 Config assembly, nominal current setting

Each channel has this parameter, which can be used to adjust the respective nominal current, which results in tripping in the event of an error.

Table 7-16 Config assembly, nominal current setting

Nominal current setting	Data type	Bit	Function	Default	Read/write
Byte 0	USINT	0 ... 7	0 = No change (EDS standard)	Depends on the module	Read/write
			1 ... 20 = 1 A to 20 A		

The respective maximum nominal current also depends on the circuit breaker module used and may vary depending on the item.

7.3.6 Config assembly, channel programming lock

This function can be used to control the nominal current setting of individual channels on the hardware side.

Table 7-17 Config assembly, channel programming

Channel current lock	Data type	Bit	Function	Default	Read/write
Byte 0	USINT	0 ... 7	0 = Lock inactive	0	Read/write
			1 = Lock active		
			2 = No change (EDS standard)		

7.3.7 Config assembly, channel status

Individual channels can be switched on and off via the channel status parameter. It is also possible to switch them back on after error-related tripping.

Table 7-18 Config assembly, channel status

Channel current lock	Data type	Bit	Function	Default	Read/write
Byte 0	USINT	0 ... 7	0 = Channel off	0	Read/write
			1 = Channel on		
			2 = No change (EDS standard)		

7.3.8 Config assembly, QUINT POWER, output voltage

Table 7-19 Config assembly, QUINT POWER, output mode

QUINT POWER, output voltage	Data type	Bit	Function	Default	Read/write
Byte 0	UDINT	0 ... 7	0 = No change (EDS standard)	2410	Read/write
Byte 1		8 ... 15	2390 2960 = 23.90 V to 29.60 V		

7.3.9 Config assembly, QUINT POWER operating mode

The QUINT POWER power supply can be operated in different modes. For a more detailed description of the respective function, refer to the operating instructions of the power supply.

Table 7-20 Config assembly, QUINT POWER, operating mode

QUINT POWER, operating mode	Data type	Bit	Function	Value range	Default	Read/write
Byte 0	UDINT	0	Parallel mode	0 = Inactive	0	Read/write
				1 = Active		
		1	Button lock	0 = Inactive	0	Read/write
				1 = Active		
		2	Default settings	0 = No reset	0	Read/write
				1 = Default setting		
		3	Shut down power supply	0 = No shutdown	0	Read/write
				1 = Shutdown		
		4 ... 7	Reserved			

7.3.10 Config assembly, QUINT POWER, relay contact

Various functions can be assigned to the relay contact of the QUINT POWER power supply.

This can be set via the following parameters.

The individual parameters of this config assembly are logically OR linked and are collectively output to the remote indication contact.

Table 7-21 Config assembly, QUINT POWER, relay contact

QUINT POWER, relay contact	Data type	Bit	Function	Value range	Default	Read/write
Byte 0	UDINT	0	DC output voltage OK	0 = Inactive 1 = Active	0	Read/write
		1	Output current	0 = Inactive 1 = Active	0	Read/write
		2	Operating hours	0 = Inactive 1 = Active	0	Read/write
		3	Temperature	0 = Inactive 1 = Active	0	Read/write
		4	Input voltage	0 = Inactive 1 = Active	0	Read/write
		5	Surge protection mode	0 = Inactive 1 = Active	0	Read/write
		6	Phase monitoring	0 = Inactive 1 = Active	0	Read/write
		7	Remaining operating hours	0 = Inactive 1 = Active	0	Read/write
Byte 1		8	Monitoring of rotating field	0 = Inactive 1 = Active	0	Read/write
		9 ... 14	Reserved			
Byte 2		15	Relay inverted	0 = Inactive 1 = Active	0	Read/write
Byte 3		16 ... 31	Reserved			

7.3.11 Config assembly, QUINT POWER, output voltage limit

Table 7-22 Config assembly, QUINT POWER, output voltage limit

QUINT POWER, output voltage limit	Data type	Bit	Function	Default	Read/write
Byte 0	UDINT	0 ... 7	25 ... 135 = 25% to 135%	90	Read/write
Byte 1		8 ... 15	90 = 90% (EDS standard)		

7.3.12 Config assembly, QUINT POWER, limit for operating hours

Table 7-23 Config assembly, QUINT POWER, limit for operating hours

QUINT POWER, limit for operating hours	Data type	Bit	Function	Default	Read/write
Byte 0	UDINT	0 ... 7	0 ... 65535 days	3650	Read/write
Byte 1		8 ... 15	3650 (EDS standard)		

7.3.13 Config assembly, QUINT POWER, limit for remaining operating hours

Table 7-24 Config assembly, QUINT POWER, limit for remaining operating hours

QUINT POWER, limit for remaining operating hours	Data type	Bit	Function	Default	Read/write
Byte 0	UDINT	0 ... 7	0 ... 5475 days	0	Read/write
Byte 1		8 ... 15	0 (EDS standard)		